

# Aniket Kumar Gupt

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## Research Interests

Flight Control & Dynamics, Dynamic Modeling and Simulation, AI/ML for Aerial Autonomy, Aerodynamics

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## Key Skills

**Software Skills:** ROS/ROS2, Gazebo, X-Plane, Planemaker, QGroundControl, MATLAB & Simulink, XFLR5, CATIA, Ansys, VS Code, Arduino IDE

**Programming Skills:** C/C++, Python

**Hardware:** ESP32, Arduino, Raspberry Pi, Pixhawk, LiDAR, MR72 RADAR, Thermal Camera, Sensor interfacing (DHT11, MQ135, IMUs, etc.)

**Flight Control, Autonomy Frameworks & AI/ML:** PX4, ArduPilot, MAVSDK, DroneKit, MAVLink, PyTorch

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## Education

**Bachelor's Degree in Aerospace Engineering, Pulchowk Campus**

2024 – 2027

IOE, Tribhuvan University, Lalitpur, Nepal

Aggregate: 82% (**Academic Excellence Award - 2024**)

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## Work Experiences

**Research Center of Low Altitude Economy Nepal**

**Flight Control Intern**

June 2025 – Present

- Assisted in system modeling, propulsion optimization, and endurance enhancement for high endurance fixed wing unmanned aerial vehicles (UAVs).
- Performed Software-in-the-Loop (SITL) and Hardware-in-the-Loop (HITL) simulations in Gazebo using PX4, followed by real-world flight validation of control algorithms.

**IITK RISE Internship**

**Research Intern**

Dec 2025

- Works on dynamic modeling and vibration analysis of gear systems, focusing on mesh stiffness variation, transmission error, and resonance behavior. Performed numerical simulations to evaluate system stability, critical speed and load deformation.

**Mach24 Orbital**

**Avionics Intern**

Oct 2024 – Nov 2024

- Assembled a 200N hybrid engine static test setup integrated actuator systems for valve control.
- Developed a pneumatic control system to operate solenoid-actuated valves, enabling reliable flow testing.

**The Squad of Changemakers**

**Mentor**

Aug 2024 – Sep 2024

- Taught foundational concepts of robotics and UAVs during the Book Free Friday initiative organized under KMC (Kathmandu Metropolitan City), promoting hands-on STEM learning.
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## Publications

Kailash Pantha, Biman Rimal, Kalyan Kumar Shrestha, David Chieng, Jeevan Koiri, Ashma Yonghang, Aniket Kumar Gupt, Anamol Khadka **An Efficient Infravision Thermal-Infrared Human Detection System for Low-Visibility Surveillance**, Feb 2026, Submitted to IEEE International Conference on ICT and photonics (ICTP2026), *Presentation Round Completed*

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## Relevant Engagements

### Flying Club, Pulchowk Campus President

May 2024 – Present

- initiated campus's first student-led aerospace and UAV initiative. Organized workshops on RC planes, autonomous drones, and flight simulation. Led the "Student Can Fly" outreach program, mentoring school students and promoting aviation awareness. Guided 100+ members in aeromodelling and led participation in national competitions like IIT Bombay Techfest, AIAA DBF fostering a culture of innovation in UAV and aerospace systems.

### AIAA Design/Build/Fly

#### Avionics and Flight Control Team

Oct 2023 – Apr 2024

#### Team Lead and Pilot

Oct 2024 – Apr 2025

#### Student Advisor and Pilot

Oct 2025 – Present

- Contributed to the selection and testing of avionics and flight control systems for a competitive UAV project. Led the team in integrating sensors, autopilot software, and control algorithms for stable and autonomous flight. Conducted sensitivity analysis, constraint analysis, and UAV performance calculations. Managed team coordination, timelines, and technical problem-solving to achieve mission objectives under strict competition constraints.

### Free Student Union, Pulchowk Campus

#### Automation Team

Aug 2025 – Present

- I contribute to designing and implementing task automation systems to improve campus operations and technical efficiency. The team focuses on IoT based automation projects that simplify routine processes and promote hands-on innovation.

### Dronacharya, LOCUS IOE Pulchowk

Feb 2025

#### 1st Runner-up

- Secured 1st Runner-up position in Dronacharya competition organized under LOCUS IOE Pulchowk by designing and piloting a drone capable of performing mission tasks.

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## Academic Projects

### Design and Development of a Quad-Motor Tilt Mechanism Fixed-Wing VTOL UAV for Autonomous Medical Payload Delivery, *Working On!!*,

- Designed a quad-motor fixed-wing VTOL UAV with dual tilt motors for efficient vertical takeoff and forward flight, optimized for autonomous medical delivery in remote regions of Nepal. Funded by CUAV and Ministry of Education, Science and Technology (MoEST), Nepal.

### Design, Fabrication, and Flight Testing of a Tricopter Tiltrotor UAV with VTOL and Fixed Wing Capabilities,

- Designed, fabricated, and tested a tricopter tiltrotor UAV with both VTOL and fixed-wing capabilities. Integrated tilt mechanisms, control systems, and conducted successful transition flight tests for hybrid operation.

### Satellite Attitude Control Simulation

- Simulated satellite attitude control in MATLAB Simulink using data from a research paper, modeling reaction wheels as DC motors with yaw angle as the control variable. Implemented and tuned a PID controller to stabilize and maintain a fixed yaw angle under constant disturbance torque.

### CubeSat Attitude Control Simulation (Python)

- Simulated 1U CubeSat attitude control using a PD controller with arbitrary gains, modeling reaction wheels, and handling disturbances. Visualized angular position, velocity, torque and wheel RPM, assuming rigid body dynamics, small-angle Euler rotations, and deterministic disturbances.

### **Autonomous Drone Navigation in GPS Denied Environments using Optical Flow, LiDAR, and Visual Sensing**

- Developed an autonomous drone navigation system for GPS denied environments with optical flow, LiDAR for positioning and visual sensing for avoidance. Implemented real-time obstacle avoidance, localization, and terrain-following capabilities for safe and intelligent UAV operation.

### **Terrain Following UAV using Digital Elevation Models (DEMs) in SITL**

- Integrated DEMs in PX4 SITL to simulate real-time terrain following, where the UAV autonomously adjusted altitude to maintain constant height above terrain in given waypoints.

### **Design and Development of a Low Cost Mobile Microscope**

TOP 8

- Funded by Ministry of Education, Science and Technology (MoEST) Nepal, this project focuses on developing a compact, affordable, and smartphone compatible microscope for use in educational purpose. Using 3D-printed parts and low cost optics, the device brings accessible microscopy to remote and resource limited areas, supporting hands on science learning and real time sample analysis.

### **Design, Fabrication and Testing of Flying Wing**

- Designed, fabricated, and tested a Flying Wing for improved aerodynamic efficiency. Focused on seamless wing-fuselage design, stability analysis, and flight testing for performance validation using Pixhawk 4.

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### **References**

**Asst. Prof. Kamal Darlami**, Deputy Head of Department, Department of Mechanical and Aerospace Engineering, Pulchowk Campus

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**Asst. Prof. Biman Rimal**, B.E. Project Coordinator, Department of Mechanical and Aerospace Engineering, Pulchowk Campus

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